LinkMate

CECS 451 Term Project, Team 10 | Phase 1: Ideation, Market Validation & Feasibility Report
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1. Industry and Target User

Industry. Personal productivity software, specifically personal relationship management (PRM): tools that help individuals, not companies, organize and use their own network.

Target users. Private individuals (B2C), defined by behaviour rather than occupation or age: they meet new people regularly in different contexts, many of these contacts are met only once or twice (weak ties), and they rely on others for advice, recommendations or help. Companies and sales teams are not targeted.

Segment	Behaviour	Role for LinkMate
Primary: Active networkers	Meet new people often and in many contexts, e.g. club members, event-goers, people who just moved to a new city	Feel the pain most; early adopters and likely paying users
Secondary: Occasional networkers	Meet new people less often but hit the same problem at specific moments	Mass market; reached via free tier and word of mouth

Situation. The problem occurs at two moments:

- **Capture:** right after meeting someone, when what the person does and offers is stored nowhere or as a loose note.
- **Need:** weeks later, when the user cannot remember who could help with a concrete problem.

2. Problem Definition

Pain point. Knowledge about contacts is fragmented across phone notes, emails, chats, LinkedIn and memory. Contact books store name, company and email, but not skills, topics discussed or help offered. Keeping notes organized takes manual effort, many small contacts are never captured, and even existing notes cannot answer “Who in my network can help with problem Y?”.

Current consequences. Users search inboxes and LinkedIn, ask around, or give up. Help that exists in their network goes unused, problems are solved more slowly or twice, referrals and collaborations are missed, and unused contacts fade over time.

Why it matters. Many needs are solved fastest by asking the right person. Research on weak ties shows that acquaintances are a key source of new information and job opportunities,[1,2] yet these are exactly the contacts whose context is lost first.

3. Proposed Solution

LinkMate is a web application that works as a personal memory for the user’s network. It can be used on a phone after meeting someone and on a laptop when searching. It combines two functions:

- **Builds contact profiles** (identity, role, skills, interests, where met, help offered) from free-text notes and enriches them over time.

[1] Granovetter, M. S. (1973). The strength of weak ties. American Journal of Sociology, 78(6), 1360–1380.

[2] Rajkumar, K., Saint-Jacques, G., Bojinov, I., Brynjolfsson, E., & Aral, S. (2022). A causal test of the strength of weak ties. Science, 377(6612), 1304–1310.

- **Finds contacts for a need** described in natural language, including indirect matches, with a short explanation and source notes for each suggestion.

Main user workflow:

1. **Capture:** the user enters a short note or pastes an email (photo and voice input are planned extensions).
2. **Review:** LinkMate extracts the profile and proposes a new contact or an update; the user confirms or corrects it.
3. **Enrich:** later interactions are added the same way, without form-filling.
4. **Ask:** the user describes a need, e.g. “Who could help me with a sensor calibration problem?”.
5. **Act:** LinkMate returns ranked contacts with reasons and sources, or states that no suitable contact exists.

4. AI Application Plan

LinkMate follows a retrieval-augmented generation (RAG) design with three AI tasks:

AI task	Input	Output	Model / algorithm
Profile extraction	Unstructured text (typed notes, pasted emails)	Structured contact profile (identity, organization, role, skills, interests, interactions), saved after user approval	Local open-source LLM (e.g. Gemma 4, Qwen3.5) via Ollama with a fixed output schema
Semantic retrieval	A need in natural language	Most relevant profile and note entries	Embedding model via Ollama; cosine similarity in Chroma
Grounded explanation	The need and retrieved entries	Ranked contacts with reason and source notes, or “no suitable contact”	Same LLM, restricted to the retrieved entries

Model selection. Local models keep third-party data on the user’s device and avoid usage costs; the final model and profile schema are chosen in the first development week.

Why AI. Notes describe skills and offers of help informally and indirectly, so rules or forms cannot capture them reliably, and manual tagging requires the effort that causes knowledge to be lost today. Users phrase needs differently from their notes, so keyword search misses contacts that semantic search finds, and rules cannot explain why a contact fits. We verify this benefit against a keyword-search baseline.

5. Initial Scope

The MVP covers the core workflow and should work reliably with about 50 contacts. The user can:

- enter a text note about a person,
- review, correct and confirm the extracted profile,
- add notes to an existing contact,
- describe a need and receive ranked contacts with explanations and source notes,
- view, edit or delete contacts in a contact list.

User flow. Input view → review form → save; chat view for needs; separate contact list.

Planned extensions. Photo and voice input, email/LinkedIn import, mobile-optimized interface.

6. Market and Existing Solutions

Product	Main focus	AI capabilities	Indicative price
Dex[3]	Personal CRM, notes, reminders	AI-assisted workflows, access via external AI tools	$12–20/month
Mesh[4] (formerly Clay)	Network built from connected accounts	Nexus assistant for network questions	Free up to 1,000 contacts; $10/month Pro
ConnectMachine[5]	Digital business cards, event capture	AI Concierge searches contacts and notes	Free plan; $5.99/month
Happenstance[6]	Warm connections across accounts	Natural-language people search with explanations	Free tier; $24/month (annual)
Covve[7]	Business-card scanning, contact management	AI contact capture and enrichment	From $12/month

AI-powered personal CRMs and natural-language contact search already exist, so LinkMate is not positioned as the first of either. The gap it addresses is **converting unstructured relationship information into continuously enriched profiles and retrieving the right person for an open-ended need, with an explanation and a traceable source for each recommendation.**

7. Business Feasibility

Value proposition. Find the right person for a problem in seconds, without manually maintaining a CRM:

- effortless capture,
- need-based search,
- preserved context,
- explainable recommendations.

Expected users. College students, entrepreneurs and freelancers who want to make better use of their existing network.

Business model. Freemium: limited contacts and text features for free; premium with unlimited contacts, cloud sync, voice input, OCR and integrations. Pricing and willingness to pay remain to be validated.

Adoption barriers and mitigation:

- **Manual entry and cold start:** minimal input, value with few contacts.
- **AI errors and privacy concerns:** review step, transparent data handling.
- **Hardware requirements:** local vs. hosted AI evaluated.
- **Competition and switching costs:** problem-oriented retrieval, easy import.

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[3] Dex. (2026). Pricing. Retrieved October 15, 2026, from https://getdex.com/pricing/

[4] Mesh. (2026). Pricing. Retrieved October 15, 2026, from https://me.sh/pricing

[5] ConnectMachine. (2026). Pricing. Retrieved October 15, 2026, from https://www.connectmachine.ai/pricing/

[6] Happenstance. (2026). Pricing. Retrieved October 15, 2026, from https://happenstance.ai/pricing

[7] Covve. (2026). Pricing — Plans for individuals & teams. Retrieved October 15, 2026, from https://covve.com/pricing

8. Technical Approach

A Python backend with a local Streamlit interface. Ollama provides a local LLM for profile extraction and explanations and an embedding model that turns profiles and notes into vectors stored in Chroma. A question is embedded, Chroma retrieves similar contact information, and the LLM generates a grounded recommendation from it. This fits the problem because contacts and needs are described informally, so semantic understanding beats fixed fields or keyword search.

9. Data Plan

- **Source and access:** no real contact data; a team-created synthetic dataset of 50–100 fictional contacts with notes (LLM-generated, checked by hand) plus 30–50 test questions labeled with the expected contacts.
- **Pipeline:** text entered in Streamlit is sent to Ollama, the proposed profile is validated and approved by the user, then saved as JSON; the LLM has no direct read/write access. Approved records are embedded in Chroma with contact and source-note IDs.
- **Grounding:** similarity thresholds, approved records only, mandatory source IDs; uncited recommendations are rejected.
- **License and privacy:** the dataset is the team's own and can be published on GitHub. In real use, data is processed locally, stays under user control (export, deletion) and is never used for training.
- **Preprocessing:** text cleaning, schema extraction, chunking and embedding.
- **Evaluation:** extraction accuracy, retrieval accuracy, citation correctness and correct refusal when no suitable contact exists.

10. Risks and Milestones

Type	Risk	Mitigation
Privacy and legal	Notes contain third-party or confidential data (GDPR, CCPA)	Local processing, deletion, synthetic data in development
Extraction quality	Wrong details, wrong contact or duplicates	User reviews every profile before saving
Hallucination	Recommendations not supported by notes	Mandatory source citations or "no suitable contact"
Prompt injection	Pasted emails manipulate the LLM	Fixed schema, no write access for the LLM
Bias	Contacts with longer notes ranked higher	Evaluation compares short and long notes
Data	Synthetic notes cleaner than real ones	Part of the test notes written by hand, incl. vague cases
Feasibility	Local models too slow or weak; scope creep	Early hardware tests, smaller models, text-only MVP
Business	Competition, capture effort, cold start	Focus on need-to-person matching and fast capture

Milestones.

Milestone	Tasks	Done when
M1 Setup and data	Define architecture; test LLMs on team laptops, choose model and schema; create synthetic dataset and test questions	Model, schema and dataset fixed
M2 PoC	Extraction into JSON profiles; embedding and retrieval via Chroma	Test questions return correct contacts
M3 MVP	Streamlit UI (input, review, chat, contact list); grounded answers with sources	Full user flow works with ~50 contacts
M4 Phase 2 delivery	First evaluation vs. keyword baseline; GitHub documentation; progress report	Runnable repository, report submitted
M5 Final evaluation and pitch	Full evaluation incl. bias check; fixes; demo video; final report	Video and final report submitted

References

Granovetter, M. S. (1973). The strength of weak ties. American Journal of Sociology, 78(6), 1360–1380.

Rajkumar, K., Saint-Jacques, G., Bojinov, I., Brynjolfsson, E., & Aral, S. (2022). A causal test of the strength of weak ties. Science, 377(6612), 1304–1310.

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Mesh. (2026). Pricing. Retrieved October 15, 2026, from https://me.sh/pricing

ConnectMachine. (2026). Pricing. Retrieved October 15, 2026, from https://www.connectmachine.ai/pricing/

Happenstance. (2026). Pricing. Retrieved October 15, 2026, from https://happenstance.ai/pricing

Covve. (2026). Pricing — Plans for individuals & teams. Retrieved October 15, 2026, from https://covve.com/pricing